

Back to Entropy: Evaluating Base Language Models Beyond Perplexity and Benchmarks

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Abstract

Base models are usually chosen from a scorecard of public benchmarks, which is an unreliable guide for two reasons: the benchmarks are public before the model exists and unstable across evaluation setups, and they score a model before adaptation, whereas teams deploy it after. Among candidates within $2\times$ of each other in size, GSM8K picks the better model for a target domain 35.9% of the time and MMLU-Pro 40.0%, below chance by point estimate. We instead adapt every candidate to recent text from the target domain under one small, identical budget and rank them by predictive loss in bits per byte, a tokenizer-agnostic measurement that costs minutes per model. Against a criterion built from each corpus’s held-out text, this ranks candidates at least as well as every public benchmark, of which only HellaSwag clears chance, across 17 models and a replication on four corpora; its lead over ranking by parameter count is statistically resolved ($+0.250$, $[+0.074, +0.500]$, lenient). The result carries to real systems: after fifteen models are fine-tuned on a news task, adapted bits per byte ranks the resulting systems correctly in 0.949 of size-matched pairs and places every model within one rank of its final position. Code and artifacts are released.

1 Introduction

A new base model ships most weeks, and everyone who might use one—a research group, a company, the team maintaining a product—asks the same question: *how good is this model, for what we do?* That is a measurement problem. What arrives with the model instead is a scorecard—MMLU-Pro, HellaSwag, GSM8K—and reading it is the obvious thing to do.

Exams are the wrong class of instrument for it, for a structural reason rather than a matter of benchmark quality. They were written by other people, for other purposes, and were public before the model existed. A high score is therefore consistent with two very different situations: the model is genuinely stronger, or the model saw more of that exam’s subject matter during pre-training. No reader can tell which, because pre-training mixtures are not disclosed. The consequence is visible inside our own cohort: Llama-3.2-1B and Qwen2.5-1.5B are 2.1 points apart on HellaSwag and 3.4% apart in adapted bits per byte, yet 55.3 points apart on GSM8K. Two models can model fresh English almost identically well and differ by almost a factor of ten on GSM8K. The scores are not even stable: identical released weights move -0.60 to $+0.69$ GSM8K points when only the *host* changes, A100 to H100, and one decoding setting a checkpoint ships with was worth 9.86 GSM8K points to Gemma-4-12B in our own re-score (Appendix G).

And none of those exams is about your text. Measured against a criterion built from a target domain’s own documents, GSM8K and MMLU-Pro select the better of two similarly-sized candidates 35.9% and 40.0% of the time by point estimate. A practitioner who reads them is not getting weak evidence about their decision; they are getting evidence about a different decision.

The second problem is timing. Every exam scores a model exactly as it ships, but no team deploys a model exactly as it ships: they adapt it to their data first. What matters is how good a model *becomes*, and a

model that looks weak out of the box—one that has simply never seen prose like yours—may need only a little adaptation to overtake one that looked stronger.

What we propose. We propose measuring the model instead: predictive loss on text the evaluator owns, after one small identical adaptation. Three properties make that work where per-token perplexity did not: *bytes* rather than tokens as the denominator, so vocabularies do not distort the comparison; *fresh* text dated after every candidate shipped, so the guarantee comes from the calendar rather than from trust; and *equal preparation*, one identical adaptation budget applied to every candidate before measuring, which turns “which model already writes like my corpus” into “which model gets there fastest.” Train-before-test [15] prepares candidates identically with labelled benchmark data; we do it unsupervised, on fresh text, and in bytes.

The minimum viable run. Text from the target domain, dated after every candidate shipped, is split 80/10/10; the smallest corpus we report on is 15 MB. Every finalist is adapted with one shared configuration—LoRA rank 16, $\alpha=32$, dropout 0.05, learning rate 10^{-4} , effective batch 32, all-linear targets, 250 steps over 512-token blocks, which is 4.1M training tokens seen per candidate—and candidates are ranked by bits per byte on the held-out test split. On one A100 this cost 3.7 GPU-minutes at 0.5B, 11.6 at 7.7B, 24.7 at 12B and 62.7 at 31B. We recommend treating a gap under 2% as a tie, roughly the measurement’s own noise floor (Section 4). The GPU time is the cheap part; assembling the corpus is the main cost, and text a product already generates needs only a clean split, not a release date.

Figure 1 is the result, and it carries through to real systems: when fifteen models are fine-tuned, adapted BPB, measured beforehand, places every one within a single rank of where it finishes. Section 2 gives the protocol and how we know the right ranking, Section 3 the measurement, and Section 4 what to do with it. Every number in this paper recomputes from a committed artifact: <https://github.com/DavidAfonsoValente/entropy-bench>.

2 The Protocol

Two things have to be built before any model can be ranked: the measurement, and the yardstick that says whether a ranking is any good. Figure 2 shows both, and neither is elaborate.

2.1 Measuring: three phases

The procedure is deliberately small enough to run on every candidate a team is considering. It has three phases, in this order, and the ordering is load-bearing: the corpus is fixed and screened before any model touches it.

Phase 1: the corpus. Collect recent text from the target domain, dated after every candidate’s release. Deduplicate exactly, filter with Gopher quality heuristics, remove MinHash/LSH near-duplicates at Jaccard 0.60, and split 80/10/10 with no identifier or normalised-text overlap across splits. For our primary snapshot—Google News, 8 June 2026, 119,054 articles—we also run three reference-free membership screens and quarantine the *union* of everything any model flags, so every candidate is measured on an identical retained corpus (Appendix B). The other three corpora (Table 4) lean on the calendar instead of a detector: Reddit and Hacker News are scraped in August 2026, after every model in the cohort shipped, and the arXiv set runs from 9 June to 20 August, so its later documents do the same work while its earliest sit alongside the news snapshot in time.

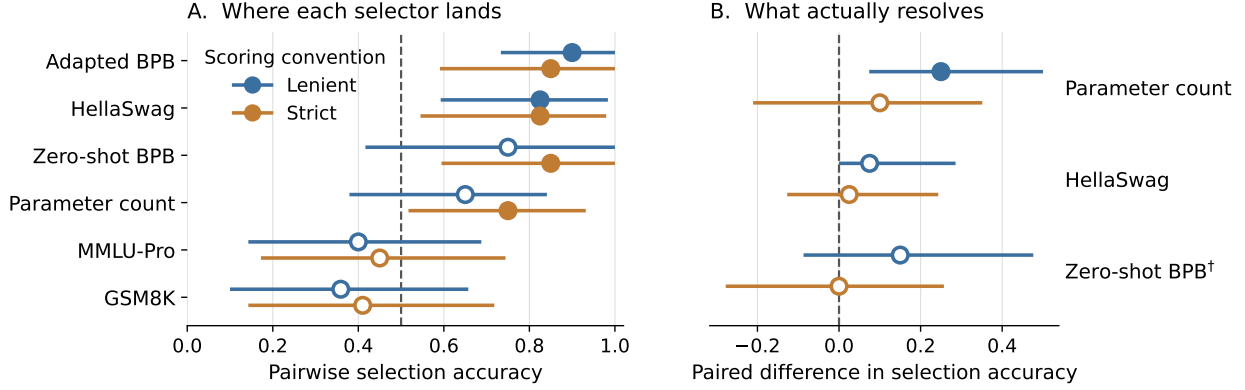


Figure 1: How often each way of choosing a base model picks the better one. *Extended cohort: 17 models, 40 size-matched pairs, news criterion.* Over the 40 pairs of models within $2\times$ in size, judged by an in-domain criterion built from the target corpus’s own held-out text (Section 2.2). **A** gives each selector’s accuracy: adapted BPB is top under lenient scoring and level with the unadapted reading under strict, and both generative benchmarks fall below chance by point estimate. **B** gives the comparison the paper actually claims—the *paired* difference between adapted BPB and each baseline, on the same pairs. Intervals are 95% cluster-bootstrap over models; a **hollow marker** means the interval still contains chance (A) or zero (B), so that comparison has not resolved. At seventeen bootstrap units exactly one has: adapted BPB over parameter count, $+0.250$ [$+0.074$, $+0.500$] lenient. Panel B plots *adapted BPB minus the named baseline*, so positive favours adaptation; the best public benchmark is HellaSwag. † computed after the fact rather than preregistered.

Phase 2: equal preparation. Pack the retained training text into contiguous 512-token blocks and adapt every candidate with the *same* configuration: LoRA [24] rank 16, $\alpha=32$, dropout 0.05, learning rate 10^{-4} , effective batch 32, all-linear targets, 250 update steps. One shared configuration rather than a per-model search is deliberate: a search we tried first handed small models more trials than large ones, a size bias no correction removes (Appendix D). One configuration suits no architecture perfectly, but we prefer a bias we can name.

Phase 3: measure in bits per byte. Compute cross-entropy on the held-out test split and divide by the UTF-8 byte length $|B|$ of the text those predictions cover:

$$\text{BPB} = \frac{\mathcal{L}_{\text{nats}}}{\ln 2} \times \frac{|\text{Tokens}|}{|B|} \quad (1)$$

The denominator is the point: a model with a larger vocabulary spends fewer tokens on the same text and looks better under per-token perplexity for free, where bytes are a physical property of the text and identical for every candidate. Test blocks are independent and non-overlapping, so context resets at every cut, and injected document-start markers stay available as context but are always masked as prediction targets. Report both endpoints, before adaptation and after, because they answer different questions (Section 3.2).

Cohort and setup. We evaluate 11 base checkpoints spanning dense Transformers, one sparse MoE and one Liquid architecture (0.5B–35B parameters; Table 12), extended to 17 in Section 3, from the Llama, Qwen, Gemma, Ministral, LiquidAI, Falcon, Mistral and SmoLLM families [25, 26, 27, 28, 29]. For comparison every model is also scored on MMLU-Pro [18], HellaSwag [19] and GSM8K [20] with *lm-evaluation-harness* 0.4.12 [12] on one host and one build, under the protocol of Appendix G—directly observed scores rather than vendor-reported card values, with the harness logs published so each recomputes from raw output. Adaptation ran on Leonardo (CINECA), 3 nodes $\times$ 4 A100 64 GB. Intervals come from one of three bootstraps, never over pairs, because pairs that share a model are not independent. Intervals on a *selector’s*

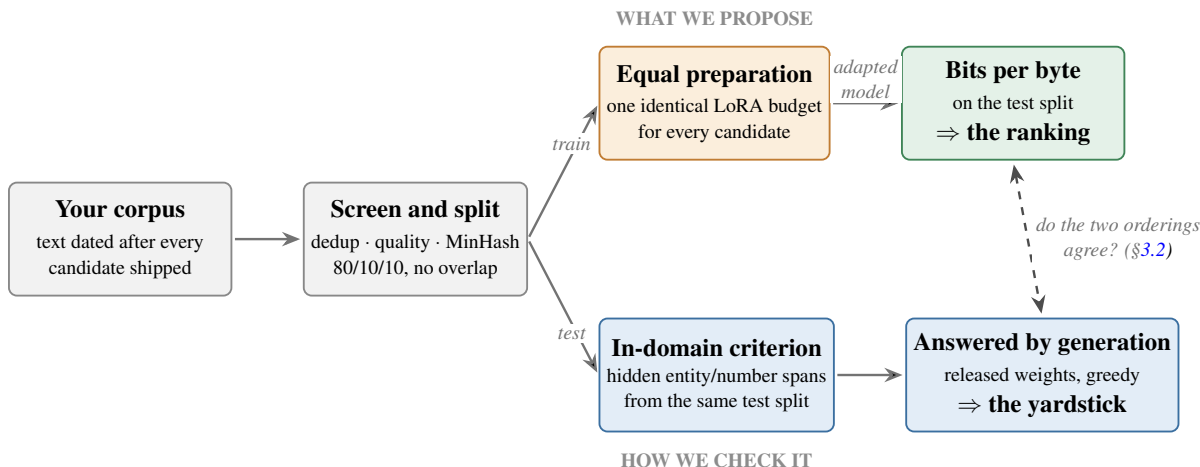


Figure 2: The whole study on one page. The upper arm is the protocol we propose: take text from your own domain that postdates every candidate, screen and split it, adapt every candidate under one identical budget, and rank by bits per byte on the held-out test split. The lower arm is how we check that ranking without appealing to the benchmarks under criticism: the *same* held-out text supplies a criterion whose items are answered by generation from the released base weights, so it is neither trainable-for nor scored by the quantity it validates. The paper is the comparison of the two orderings.

accuracy resample **models** (20,000 draws, percentile, clustered) and ask whether the selector would still win on another draw of models. Intervals on the *criterion’s own* scores resample its 2,500 **items** instead, paired across models, and ask how well the yardstick itself is known (Section 2.2); intervals on a fine-tuned pair resample its 500 evaluation articles. Each is named where it is used.

2.2 How we know the right ranking

To show one way of ranking models beats another, we first need to know which models really are better. There is no universal answer—a model is only ever better *at something*—so we ask the question a team asks: *which model is better at the job, in this domain?* We answer it two independent ways.

Fine-tune them and see. The most direct test is to do exactly what a team does: fine-tune every model on a real task and score the systems that come out (Section 3.4). The model that builds the best system is the better model. It is expensive, so we run it once, on fifteen models.

A quiz on articles no model has seen. The cheap, repeatable version is a fill-in-the-blank quiz built from the domain’s own text. We take held-out articles no model was adapted on, hide a name or a number, show the model the 1,200 characters before it, and ask it to write the missing words; a model’s score is simply how many it gets right (Appendix B shows one). We build one quiz per corpus, four in all, and call the ranking it produces *the criterion*.

The two agree. On news, the corpus where we ran both, the quiz ranking and the fine-tuned ranking of the fifteen models correlate at Spearman 0.921 (lenient scoring). That check is what lets the cheap quiz stand in for the expensive fine-tune, and it is the yardstick every result below is judged against.

Quiz articles come from each corpus’s held-out test split. Reddit and Hacker News postdate every model’s release, as does most of the arXiv set; the news snapshot predates part of the release window and is protected by the screen of Phase 1 instead (Table 4). A hidden span that already appears earlier in the prompt is excluded, so a right answer means knowing, not copying, and answers are generated and matched as strings. Three properties make the quiz a fair judge:

- **It cannot have been prepared for.** For the three scraped corpora the guarantee is the calendar rather than

trust—Reddit and Hacker News did not exist when any of the weights were frozen, and most of the arXiv set did not either—and for the news snapshot it is the membership screen.

- **It is not scored by likelihood.** HellaSwag ranks continuations by length-normalised conditional log-likelihood—the same object BPB measures—so agreement between BPB and any likelihood-scored criterion would be convergent validity rather than evidence. Generation and string match put distance between the two, though greedy decoding is still the argmax of the same conditional.
- **It is answered by the released weights, not the adapted ones,** so it is independent of the very adaptation whose BPB it validates. This is the load-bearing property of the three: whatever the criterion and BPB share, they cannot share the adaptation under test.

One detail matters for reading every number that follows. A base model has no stop convention and will run on past the answer, so exact-prefix matching partly measures stopping behaviour rather than knowledge. We therefore score every item both ways—*strict*, the answer as a prefix of the generation, and *lenient*, the answer anywhere in a 24-character window—and require every *ordering* against a public benchmark to hold under both—and adapted BPB never falls behind a public benchmark under either (Table 2).

3 Results

Which models carry which claim. Two cohorts run through this section and it is worth fixing them before the first number. The headline selector comparison (Section 3.1, Figure 1) uses the **extended 17-model cohort** and its 40 size-matched pairs, judged against the news criterion. Everything after it—the mechanism, the four-corpus replication and the stability results—uses the **original 11-model cohort**, the one for which all four corpora and both BPB endpoints exist for every model; there it yields 10–11 size-matched pairs per corpus, and 55 pairs when the size band is dropped. Each table names its own population.

3.1 Adapted BPB ranks candidates better than any benchmark we tested

The regime that matters most is candidates within $2\times$ of each other in size. Outside it, size does most of the work—yet adapted BPB still beats it across all 136 pairs ($+0.088$ [$+0.016, +0.167$], lenient). Inside it, the choice is genuinely hard—and eleven models yield only eleven such pairs, so we extended the cohort to 17 on two preregistered criteria and no others, size-adjacency and publisher spread (three checkpoints were excluded on infrastructure grounds, never for their scores; Table 11 lists the cohort and Appendix F the rules). That gives **40** size-matched pairs, and the result does not lean on any one publisher: 32 of the 40 pairs cross publisher families, and adapted BPB is right on 30 of them (0.938 lenient, 0.875 strict).

Figure 1 is the paper’s central measurement, and the same ordering holds on real fine-tuned systems, ranked correctly in 0.949 of pairs (Section 3.4). Adapted BPB selects the better candidate 0.900 of the time under lenient scoring, ahead of every alternative, with HellaSwag next and the unadapted reading and parameter count behind it. Under strict scoring it is 0.850 and the unadapted reading is level with it; no alternative is ahead under either convention. Both generative benchmarks land *below* chance by point estimate, GSM8K at 0.359. Their intervals still contain chance at this cohort size, so what is established is that neither is usable in this band—not that either is reliably worse than a coin.

The lead over ranking by size is statistically resolved: $+0.250$, 95% interval [$+0.074, +0.500$], positive in 99.4% of resampled cohorts under lenient scoring ($+0.100$ under strict). Against HellaSwag, the strongest public benchmark, adapted BPB holds the top rank and never falls behind it on any corpus—while measuring your own text rather than a fixed public exam. At forty pairs **HellaSwag clears chance** too—which an eleven-model cohort was too low-powered to show—and a larger cohort is what will resolve the gap between the two.

3.2 Why it works: adaptation is scoring capability instead of familiarity

The mechanism is specific and checkable.

Table 1 puts each model’s rank under the independent criterion beside its rank under the two BPB readings. Ranking by the adapted number reproduces the criterion’s ordering almost exactly: seven of eleven models land in precisely the right position and the mean rank error is 0.36 places. The unadapted reading places two of eleven correctly and is off by 2.00 places. Eleven of the fifty-five model pairs flip from wrongly to correctly ordered when the reading changes, and none flips the other way. That gap is not an artefact of a noisy yardstick: resampling the criterion’s own items while holding both BPB orderings fixed leaves the adapted reading at 0.36 [0.00, 0.55] places against 2.00 [2.00, 2.18] for the unadapted one, and the difference between them, 1.64 [1.45, 2.00], excludes zero.

What the unadapted number is actually measuring. Zero-shot loss conflates two things a practitioner needs separated: *this model already writes like my corpus* and *this model is a capable model*. Figure 3 shows the confound in the raw loss, resolved by position inside a packed evaluation block. Because blocks are independent, every block is a cold start. Unadapted, Gemma-4-12B opens a block at 2.83 bits per byte where Qwen-2.5-1.5B—eight times smaller—opens at 2.04, and it stays behind for roughly the first half of the block; by the end of the block it has passed it, 0.72 against 0.77. A block-averaged number therefore reports mostly how quickly a model finds its footing in unfamiliar prose, which is a real property but not the one the decision turns on. A short equal adaptation supplies exactly the missing familiarity, and what is left is closer to capability.

The prediction this makes, and it holds. If that is the mechanism, then the models the unadapted reading most under-rates must be the ones adaptation helps most. They are: the three largest zero-shot-to-adapted reductions in the cohort are Gemma-4-12B, Gemma-4-31B and LFM2.5-1.2B, which are also the three the free reading most under-rates. Across the cohort the correlation between a model’s adaptation gain and its zero-shot rank error is 0.672; against its *adapted* rank error it is 0.024. The confound shows up in the unadapted number and not in the adapted one.

Scoring. Lenient scoring is the primary convention for the reason given in Section 2.2; under strict the two readings are level (0.850 each on the 17-model cohort, 0.727 on the eleven).

The cold start is part of the story, not all of it. If it were the whole story, refusing to score the opening should recover the adapted reading—and it does not. Scoring every model zero-shot while discarding the first K positions of each block lifts in-band accuracy from 0.545 to at best 0.636 at $K=128$, a quarter of the way to 0.909, the adapted reading’s score on the same eleven models. That also answers the cheaper reading of Table 1—that the criterion scores a warmed-up regime, adapted BPB is warm too, and the two merely meet there—because warming the free reading did not close the gap (Appendix D).

Two cases worth naming. **Gemma-4-12B** is ninth of eleven on zero-shot BPB but second on both the independent criterion and adapted BPB—a 12B model the free reading puts below two models under 2B. **LFM2.5-1.2B** is the same effect at the other end: dead last zero-shot at 1.34 bits per byte against a 0.5B Qwen’s 0.93, yet ahead of it after adaptation and on the criterion—a non-Transformer penalised hardest by a metric that rewards familiarity.

Model	Independent criterion			Rank by	
	score	95% CI	rank	zero-shot BPB	adapted BPB
Gemma-4-31B	0.213	[0.197, 0.229]	1	4	1
Gemma-4-12B	0.186	[0.171, 0.201]	2	9	2
Qwen-3.5-35B-MoE	0.147	[0.134, 0.161]	3	1	3
Ministral-3-14B	0.142	[0.128, 0.155]	4	2	4
Qwen-2.5-7B	0.121	[0.108, 0.134]	5	5	6
Qwen-3.5-9B	0.116	[0.104, 0.129]	6	3	5
Llama-3.2-1B	0.104	[0.092, 0.116]	7	7	8
Qwen-3.5-4B	0.101	[0.089, 0.112]	8	6	7
Qwen-2.5-1.5B	0.089	[0.078, 0.100]	9	8	9
LiquidAI-LFM2.5	0.078	[0.067, 0.088]	10	11	10
Qwen-2.5-0.5B	0.066	[0.057, 0.076]	11	10	11
<i>mean rank error</i>				2.00	0.36
<i>ranks exactly right</i>				2 of 11	7 of 11

Table 1: What the adaptation step buys. The eleven-model cohort on the *news* corpus, ordered by an in-domain criterion built from that corpus’s own held-out text and answered by the released base weights. Ranking by adapted BPB reproduces that ordering almost exactly; ranking by the unadapted reading does not. Bold marks an exact match with the criterion. Gemma-4-12B is the clean case: ninth by the free number, second by the criterion, second by the adapted number. Scores are lenient. Intervals are a paired bootstrap over the 2,500 shared cloze items; three of the ten adjacent pairs—Qwen-3.5-35B-MoE over Ministral-3-14B, Qwen-2.5-7B over Qwen-3.5-9B, and Llama-3.2-1B over Qwen-3.5-4B—are not separated by the criterion at all, and nothing in this paper turns on their order. For scale, these are mixed entity-and-number items: within the news corpus numeric spans are answered at 0.279 and entity spans at 0.180 (Appendix B), so 0.213 is a strong score on this set rather than a weak one on an easy one.

3.3 The same comparison on four independent yardsticks

One criterion is one opinion. Table 2 runs the entire comparison four times, each corpus judged only against a criterion built from its own held-out text: news, Reddit, Hacker News and arXiv mathematics. Two patterns hold across all four domains and both scoring conventions: adapted BPB never falls below a public benchmark in any row, and no public benchmark reaches the top of any row. Against every selector, size included, it leads or ties five of the eight. On news and Reddit both generative suites sit far below chance by point estimate—0.182 on the news criterion, picking the worse model four times out of five—while on Hacker News MMLU-Pro reaches 0.727 and on mathematics both recover to 0.700–0.800.

Ranking by size scores higher in three of the rows—news strict and both Hacker News rows, at 0.818, 1.000 and 0.818. Size is a strong baseline, and adapted BPB still beats it across the full range of all 136 pairs.

3.4 What it changes

Aggregates hide the thing a practitioner wants to see: which model does this actually change (Table 7 prices each choice)? Gemma-4-12B is the case to look at. Among the eleven models of Table 8 it is second by adapted BPB, second by the in-domain criterion, and *seventh* on both MMLU-Pro and GSM8K. A team reading the scorecard would not shortlist it. Scorecard gaps are also wildly out of scale with what they decide, as the fine-tune below shows.

So we fine-tuned them. Every criterion so far lives in the corpus the selector adapted on, so we left it. We fine-tuned fifteen of the seventeen models—every one that fits a single GPU—on a real in-domain task,

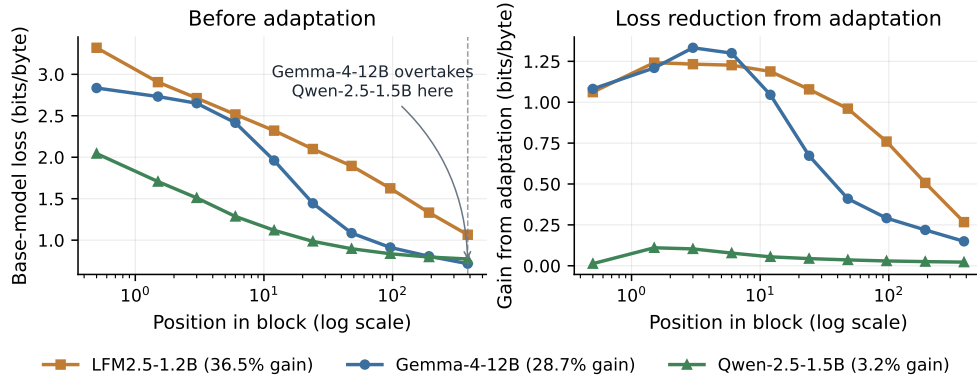


Figure 3: Why the unadapted number mis-ranks: it is largely scoring cold starts. Base loss by position within a packed 512-token evaluation block. Unadapted, Gemma-4-12B opens a block at 2.83 bits per byte against Qwen-2.5-1.5B’s 2.04—a 12B model losing to a 1.5B one—and closes it at 0.72 against 0.77. A block average over-weights the opening, and that is *part* of what puts Gemma-4-12B ninth of eleven on zero-shot BPB (Table 1): discarding the opening outright recovers about a quarter of the gap between the two readings and no more (Appendix D). *Right*: the loss reduction adaptation buys at each position is largest in the opening but does not vanish after it—adaptation teaches the corpus’s opening conventions, and does more besides.

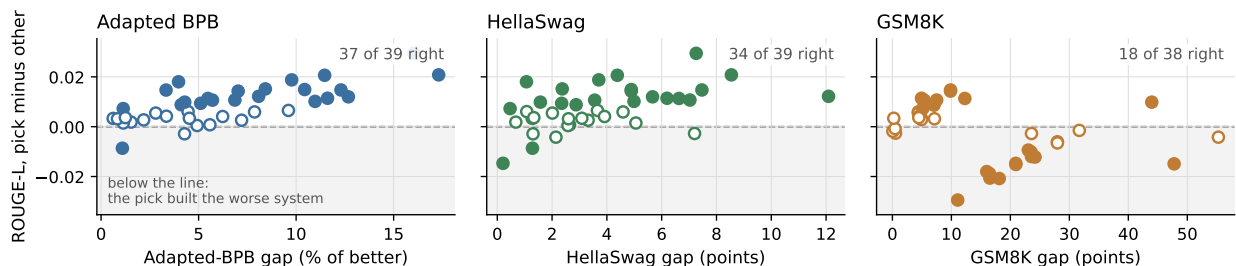


Figure 4: Did the selector’s pick build the better system? Every size-matched pair of the fifteen fine-tuned models, oriented so the selector prefers the first: x is how strongly it prefers it, y how much better that model’s fine-tuned system scored. Above the line is a pair the selector got right; filled markers are pairs the fine-tune itself resolves. Adapted BPB is right on 37 of the 39 pairs, and on 21 of the 22 the fine-tune separates; HellaSwag follows at 34. GSM8K is right on 18 of 38, and every one of its eleven gaps between 15 and 25 points sits *below* the line, on a pair where it picked the worse system.

writing the opening paragraph of a news article from its body, with one identical LoRA recipe and ROUGE-L against the human-written lede on 500 held-out articles from a split neither the criterion nor the BPB test blocks touch. The analysis was fixed in code before the jobs ran (Appendix E).

Scored against the fine-tuned systems over all 39 size-matched pairs they form, the headline repeats on an outcome nothing here was tuned against. Adapted BPB orders 0.949 of the pairs correctly, $[0.817, 1.000]$, ahead of HellaSwag at 0.872, while GSM8K and MMLU-Pro sit at 0.474 and 0.513—a coin (Figure 4, Table 10). Its lead over parameter count is statistically resolved under this recipe, $+0.231 [+0.059, +0.452]$, and on the pairs the fine-tune itself separates it picks the better system in 21 of the 22. A deliberately different recipe—four times the rank, three times the learning rate, half the batch, a new seed—leaves the systems’ ordering almost unchanged (Spearman 0.982) and adapted BPB still first at 0.923, right on all 20 pairs that recipe separates. Table 3 shows why this matters: a model’s benchmark standing is a poor guide to where it finishes, while adapted BPB places all fifteen within one rank of their final order. **Gemma-4-12B**, 9th and 8th of fifteen on GSM8K and MMLU-Pro, builds the best system of the fifteen, beating Ministral-3-14B by $+0.018 [+0.011, +0.025]$; and the 55.3-point GSM8K gap between Llama-3.2-1B and Qwen-2.5-1.5B buys no measurable difference once both are fine-tuned. Every model improved on its own base weights

Criterion built from	scoring	Adapted BPB	Zero-shot BPB	HellaSwag	GSM8K	MMLU-Pro	Param. count
Google News (11)	lenient	0.909 [†]	0.545	0.818	0.182	0.182	0.636
	strict	0.727	0.727	0.636	0.364	0.364	0.818
Reddit (11)	lenient	1.000	0.636	0.909	0.273	0.273	0.727
	strict	0.909	0.727	0.818	0.364	0.364	0.818
Hacker News (10) (11)	lenient	0.900	0.800	0.800	0.500	0.600	1.000
	strict	0.727	0.818	0.636	0.545	0.727	0.818
arXiv math (10)	lenient	0.900	0.900	0.600	0.700	0.700	0.800
	strict	1.000	1.000	0.600	0.800	0.800	0.800

Table 2: The same comparison, run four times against four independent yardsticks. *Eleven-model cohort; 10–11 size-matched pairs per corpus.* Each row block scores every selector against a criterion built from *that* corpus’s own held-out text—post-release for Reddit, Hacker News and arXiv, screened rather than post-dated for news (Section 2.2)—over the size-matched pairs (count in parentheses; Hacker News has ten under lenient scoring and eleven under strict, because a pair the criterion ties orders nothing). Bold is the leader in the row; † marks an interval clearing chance. Two things replicate across all four domains and both conventions: adapted BPB never falls below a public benchmark in any row, and no public benchmark reaches the top of any row; against every selector it leads or ties five of the eight. It holds the highest point estimate on two of the four corpora under both conventions—outright on Reddit, tied with the unadapted reading on arXiv—while ranking by size leads three rows. At ten or eleven pairs per corpus almost no cell here has the power to clear chance—one of the forty-eight does—so this table is read for its orderings, not its intervals. GSM8K and MMLU-Pro fall as low as 0.182.

Model	Out of the box			After	Adapted BPB
	raw loss	GSM8K	MMLU-Pro	<i>fine-tuning</i>	<i>predicted</i>
Gemma-4-12B	13	9	8	1	2
Llama-3.2-1B	10	15	15	11	11
LFM2.5-1.2B	15	12	13	14	14

Table 3: How a model looks is not how it finishes. Rank among the fifteen fine-tuned models (1 is best). The benchmarks misjudge where these models end up, and raw loss misjudges Gemma-4-12B by twelve places: it is mid-table on both benchmarks and near the bottom on raw loss, then builds the best system of all. Llama-3.2-1B is last on both benchmarks and finishes eleventh. Adapted BPB, measured before any task fine-tuning, places every model within a single rank of where it finishes—all fifteen, and eleven exactly.

and cleared what copying the article’s first sentence scores (0.125).

4 Recommended Practice

1. **When candidates differ by more than about $2\times$ in parameters, the larger one is usually the better choice.** Size is a good rule of thumb there, though adapted BPB beats it even across the full $70\times$ range; the protocol is built for the case where size has stopped deciding.
2. **The corpus should come from the target domain, or close to it,** dated after every candidate shipped and screened as in Section 2. Nearby text can work as well: on the ten pairs we tested, Reddit BPB predicted the news fine-tune as well as news BPB did (Appendix E). The choice of corpus determines what the ranking tracks, so it should match the capability that matters.
3. **Every finalist is adapted under one identical budget and ranked by adapted BPB,** at the cost in GPU-minutes given on page 3: cheap enough to run on every candidate rather than a shortlist.

4. **The unadapted number is best read as a free screen, not the selector.** A candidate badly off-distribution for the target text shows up in it, but on its own it reaches only 0.545 against the in-domain criterion in the size-matched band under lenient scoring, with an interval spanning chance; under strict the two readings are level, so this is the one place adaptation may buy nothing.
5. **With two or three candidates rather than a cohort, the gap is what matters, and we recommend treating anything under 2% as a tie.** Two floors say what a gap must clear: re-running the *same* model on a different seed moved adapted BPB by up to 1.24% (Gemma-4-12B; the median model, 0.01%), and token-aligned blocks add up to 0.18% (Appendix D); the worst case, rounded up, is 2%. Above it: of the 33 size-matched pairs at least 2% apart, the lower-BPB model was the criterion’s preference in **all** 33 lenient and 30 of 33 strict. Closer than that, the choice is better made on licence, latency or serving cost. The threshold was chosen after seeing this cohort, so Table 9 is best read as a calibration rather than an estimate.

Results are comparable only when corpus revision, split, context length, marker masking, adaptation configuration and update budget all match; changing any of them creates a new benchmark rather than a new measurement of the old one.

5 Conclusion

Public benchmarks were built to compare models in general, and that makes them an unreliable guide to the narrower question most teams face: which base model will serve a particular domain best. This work shows that a cheap measurement answers that question better. Adapting every candidate to recent in-domain text under one identical, small budget and ranking by bits per byte selected the better model at least as often as any public benchmark across four corpora, resolved its lead over parameter count under lenient scoring, and predicted the quality of fifteen real fine-tuned systems, placing each within one rank of its final position.

Why it matters. The measurement is tokenizer-agnostic, costs minutes of GPU time per candidate, and is computed on text the evaluator controls, so it cannot be prepared for in advance and can be rerun whenever a new model appears. It also captures what a benchmark structurally cannot: how good a model becomes once adapted, which is the quantity a deployment depends on. For research it offers a contamination-resistant complement to public leaderboards, and a way to compare models that does not rest on any fixed exam.

Anticipated objections. Four concerns arise naturally, and each was tested directly. *Circularity*—that a yardstick built from the same corpus rewards the method—is answered by fine-tuning the models and scoring the resulting systems on separate documents against human-written references; that ranking agrees with the criterion at Spearman 0.921. *Recipe dependence*—that the downstream agreement reflects the adaptation recipe—is answered by repeating every fine-tune with a substantially different recipe, which leaves the ranking nearly unchanged (Spearman 0.982). *Contamination* is addressed with text dated after the models’ release or screened for membership, and *selection bias* by scoring every size-matched pair under preregistered rules. The main open question is statistical power: with seventeen models the lead over HellaSwag is consistent but not yet resolved, and a larger, more diverse cohort is the natural next step.

References

- [1] Shannon, C. E. (1951). Prediction and entropy of printed English. *Bell System Technical Journal*, 30:50–64.
- [2] Mielke, S. J. (2019). Can you compare perplexity across different segmentations? Technical note.
- [3] Gao, L., et al. (2021). The Pile: An 800GB Dataset of Diverse Text for Language Modeling. *arXiv:2101.00027*.

- [4] Ahia, O., et al. (2023). Do All Languages Cost the Same? Tokenization in the Era of Commercial Language Models. EMNLP. arXiv:2305.13707.
- [5] Magnusson, I., et al. (2023). Paloma: A Benchmark for Evaluating Language Model Fit. arXiv:2312.10523.
- [6] Carlini, N., et al. (2021). Extracting Training Data from Large Language Models. USENIX Security.
- [7] Carlini, N., et al. (2022). Quantifying Memorization Across Neural Language Models. ICLR 2023. arXiv:2202.07646.
- [8] Lee, K., et al. (2022). Deduplicating Training Data Makes Language Models Better. ACL. arXiv:2107.06499.
- [9] Shi, W., et al. (2023). Detecting Pretraining Data from Large Language Models. ICLR 2024. arXiv:2310.16789.
- [10] Zhang, J., et al. (2024). Min-K%++: Improved Baseline for Detecting Pre-Training Data from Large Language Models. ICLR 2025. arXiv:2404.02936.
- [11] Xu, C., et al. (2024). Benchmark Data Contamination of Large Language Models: A Survey. arXiv:2406.04244.
- [12] Gao, L., Biderman, S., et al. (2024). A Framework for Few-Shot Language Model Evaluation (lm-evaluation-harness). EleutherAI.
- [13] Heineman, D., Hofmann, V., Magnusson, I., et al. (2025). Signal and Noise: A Framework for Reducing Uncertainty in Language Model Evaluation. NeurIPS 2025.
- [14] Thrush, T., Potts, C., and Hashimoto, T. (2024). Improving Pretraining Data Using Perplexity Correlations. arXiv:2409.05816.
- [15] Zhang, G., Dominguez-Olmedo, R., and Hardt, M. (2025). Train-before-Test Harmonizes Language Model Rankings. arXiv:2507.05195.
- [16] Ruan, Y., Maddison, C. J., and Hashimoto, T. (2024). Observational Scaling Laws and the Predictability of Language Model Performance. NeurIPS 2024.
- [17] Gadre, S. Y., Smyrnis, G., Shankar, V., et al. (2025). Language Models Scale Reliably with Over-Training and on Downstream Tasks. ICLR 2025.
- [18] Wang, Y., et al. (2024). MMLU-Pro: A More Robust and Challenging Multi-Task Language Understanding Benchmark. NeurIPS D&B. arXiv:2406.01574.
- [19] Zellers, R., et al. (2019). HellaSwag: Can a Machine Really Finish Your Sentence? ACL. arXiv:1905.07830.
- [20] Cobbe, K., et al. (2021). Training Verifiers to Solve Math Word Problems (GSM8K). arXiv:2110.14168.
- [21] Krause, B., Kahembwe, E., Murray, I., and Renals, S. (2018). Dynamic Evaluation of Neural Sequence Models. ICML 2018. arXiv:1709.07432.
- [22] Krause, B., Kahembwe, E., Murray, I., and Renals, S. (2019). Dynamic Evaluation of Transformer Language Models. arXiv:1904.08378.
- [23] Gururangan, S., et al. (2020). Don’t Stop Pretraining: Adapt Language Models to Domains and Tasks. ACL. arXiv:2004.10964.
- [24] Hu, E. J., et al. (2021). LoRA: Low-Rank Adaptation of Large Language Models. ICLR 2022. arXiv:2106.09685.
- [25] Dubey, A., et al. (2024). The Llama 3 Herd of Models. arXiv:2407.21783.
- [26] Yang, A., et al. (2024). Qwen2.5 Technical Report. arXiv:2412.15115.
- [27] Gemma Team (2026). Gemma 4 Technical Report. arXiv:2607.02770.
- [28] Liu, A. H., et al. (2026). Ministral 3. arXiv:2601.08584.
- [29] Qwen Team (2026). Qwen3.5: Towards Native Multimodal Agents. Official release and model card. <https://qwen.ai/blog?id=qwen3.5>.

A Related Work

From perplexity to comparable loss. Language modelling was grounded in byte- and character-level entropy from the start [1], but token-level perplexity depends on each model’s vocabulary [2, 4]; normalising by UTF-8 bytes restores a common unit [3, 5], and loss-based metrics carry better signal and lower noise than accuracy across 465 models [13]. Our contribution is the operational protocol around the metric, not the metric.

Where the loss is measured matters. Thrush et al. turn the varying correlation between per-domain perplexity and benchmark performance into a pretraining-data selection rule [14]. Their estimand is the complement of ours: they fix the benchmark suite and choose data to *train* one strong model, where we fix the candidate pool and ask how the *selection ranking* moves.

Equal preparation. Dynamic evaluation updates a model on recent test history [21, 22] and domain-adaptive

pretraining improves fit to a target distribution [23]. Closest to our motivation, *train-before-test* gives 61 models identical supervised preparation across 24 tasks and raises cross-benchmark Kendall agreement from 0.52 to 0.76 [15]; they rank post-fine-tuning accuracy on public labelled benchmarks, where we rank held-out fit after *unsupervised* adaptation to a fresh corpus, across tokenizers. The core idea—prepare every candidate identically before comparing them—is theirs; what we add is doing it without labels, on text no vendor has seen, in a unit that compares tokenizers, and checking the result against real fine-tuned systems. Controlled families show a smooth loss-to-error relation [17, 16]; we test it *across* families.

Contamination. Models memorise duplicated pretraining text [6, 7], so deduplication and contamination analysis are necessary [8, 11]; reference-free membership signals are useful screens [9, 10] but prove nothing about a proprietary training set, so we use them only for conservative quarantine.

Corpus	Raw records	Documents	Median words	Text	Contamination control
Google News, 8 Jun 2026	119,054	119,054	362	361 MB	Min-K++ / context / perturbation union
Reddit, 6–10 Aug 2026 [†]	3,054,623	141,527	624	533 MB	Post-release scrape
Hacker News, 6–10 Aug 2026	60,814	6,086	690	46 MB	Post-release scrape
arXiv math. *, 9 Jun–20 Aug 2026 [‡]	3,207	3,207	—	15 MB	Post-release scrape + GSM8K shingle screen

Table 4: Corpora. All four receive identical exact-deduplication, Gopher quality filtering and MinHash/LSH near-duplicate removal at Jaccard 0.60; they differ in *contamination* control, deliberately. The reference-free screen ran on the news corpus, whose 8 June 2026 snapshot predates part of the cohort’s release window; Reddit and Hacker News were scraped after every model shipped, as was most of the arXiv set, and they are protected by construction. Reddit and Hacker News comments are grouped to document length. [†]Four days are available in the stated range. [‡]Used only for the mathematics adaptation.

B The Criterion: Construction and Composition

Table 5 gives the four criteria. Items come from each corpus’s own held-out test split, reproduced with the evaluation pipeline’s shuffle and seed. Each hides one entity or number span and supplies 1,200 characters of preceding context; spans occurring earlier in the prompt are dropped, so retrieval from context cannot substitute for prediction. Generation is greedy, scored by string match, and produced by the released base weights.

Composition differs across the four, which is why we never compare their levels: news is entity-dominated and Reddit number-dominated, and numeric spans are simply easier—answered at 0.279 and 0.325 against 0.180 and 0.060 for entity spans on those two corpora. Within a corpus every selector is scored on identical items, which is the only comparison the paper draws.

A worked item, answered by two of the cohort:

... MBDA is owned by Airbus (37.5%), BAE Systems (37.5%), and Leonardo (____
hidden span: 25% Gemma-4-12B: 25%) . ✓ *Qwen-2.5-0.5B: 30%)* . ×

The answer is nowhere in the context; the model has to know the company, or do the sum.

How precisely is the quiz known? We resample its 2,500 items, paired across models so that item difficulty cancels out of a between-model difference. It separates the cohort from 0.066 to 0.213 and resolves seven of the ten adjacent pairs under lenient scoring, two under strict; Table 1 carries every interval. Adjacent models it cannot order are reported as unordered rather than counted as some selector’s mistake.

Does sharing text with BPB make it circular? The criterion and the BPB evaluation do come from the same held-out bytes. The evidence that this does not drive the result is internal: the reading *mechanically closest* to the criterion—zero-shot BPB, same weights, same text, same conditional—is the weaker of the two at 0.545, while the adapted reading, from different weights, is the best. Shared bytes alone do not buy agreement, and Section 3.4 leaves the loop altogether: a real fine-tune, scored on different documents against a human-written reference.

Is it just measuring size? Such a criterion could also be a size proxy. Over the full $70\times$ range it does track size ($\rho = 0.918$ against log parameters), as any capability measure should. *Inside* the band, parameter count clears chance against none of the four per-corpus criteria, though it does on the extended news cohort under strict (0.750 [0.517, 0.932]), and two dense size inversions resolve—Gemma-4-12B over the larger Ministral-3-14B, +0.044 [+0.032, +0.057], and Llama-3.2-1B over Qwen-2.5-1.5B, +0.016 [+0.006, +0.026]. (The one mixture-of-experts model activates 3B of 34.7B and is never used as a size example.)

Screening. The news corpus additionally passes three reference-free membership screens per model—Min-K++ (normalised likelihood of the lowest-probability $K=20\%$ target tokens), context sensitivity (target NLL with and without sampled same-corpus prefixes), and, for flagged candidates only, preference over two word-swap/placeholder perturbations. The global union quarantined 80 unique sequences, after which every model receives the identical retained corpus. Near-100% preference for the unperturbed original is expected—fluency alone produces it—so we draw no memorisation conclusion from it (results/contamination/).

Criterion	Items	Entity	Number	Drawn from
Google News, 8 Jun 2026	2,500	1,677	823	test split of a 119,054-doc corpus
Reddit, 6–10 Aug 2026	2,500	900	1,600	test split of a 141,527-doc corpus
Hacker News, 6–10 Aug 2026	2,005	931	1,074	test split of a 6,086-doc corpus
arXiv math. *, 2026	1,911	949	962	test split of a 3,207-doc corpus

Table 5: The four criteria. Composition differs by corpus and that is why their *levels* are not comparable across rows: news is entity-dominated, Reddit number-dominated. Within a corpus the comparison is between selectors on identical items, which is the only comparison we make. The mathematics set is the thinnest, drawing on 320 held-out documents with up to eight spans from each.

C The Full Benchmark Alignment

Table 6 reports all twenty-four (corpus $\times$ reading $\times$ benchmark) cells over the full 55 pairs; Table 7 prices each reading’s top-1 pick against each benchmark’s own. The rotation (Appendix D) is legible column by column: adapting on general text raises HellaSwag agreement and lowers the GSM8K column on every corpus and MMLU-Pro on two of the three general-text corpora; adapting on mathematics reverses the GSM8K half. The pairs-correct column—GSM8K only—shows how little of this is visible in a rank correlation: across its eight rows it spans 41/55 to 49/55.

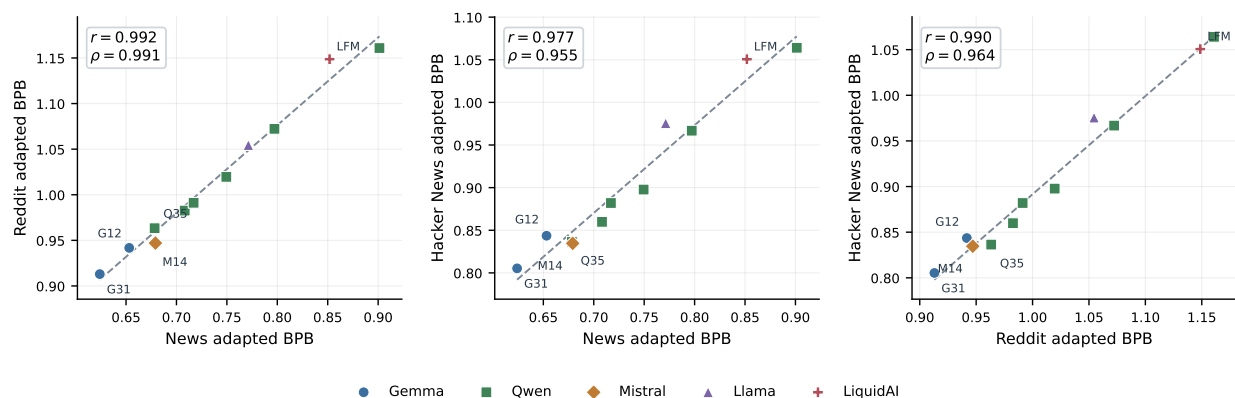


Figure 5: Pairwise adapted-BPB agreement across domains. Absolute difficulty changes, but points remain close to a monotone shared ordering: Pearson correlations between score values are 0.992 (news–Reddit), 0.977 (news–Hacker News) and 0.990 (Reddit–Hacker News). G31/G12 denote Gemma-4-31B/12B; Q35 is Qwen-3.5-35B-MoE; M14 is Ministral-3-14B.

Corpus	Reading	Rank correlation (ρ) with			GSM8K pairs
		GSM8K	MMLU-Pro	HellaSwag	correct
News	zero-shot	0.800	0.845	0.718	45/55
	adapted	0.700	0.773	0.982	42/55
Reddit	zero-shot	0.745	0.782	0.655	43/55
	adapted	0.655	0.745	0.964	41/55
Hacker News	zero-shot	0.800	0.809	0.627	44/55
	adapted	0.745	0.855	0.973	44/55
Math (arXiv)	zero-shot	0.764	0.736	0.482	43/55
	adapted	0.909	0.927	0.818	49/55

Table 6: *Eleven-model cohort, all 55 pairs (no size band).* All twenty-four (corpus $\times$ reading $\times$ benchmark) cells, recomputed from the committed cells. The rotation is visible column by column: adapting on general text raises HellaSwag agreement and lowers GSM8K everywhere, while adapting on mathematics reverses it.

Corpus and reading	Picks	Accuracy given up vs. the benchmark’s own pick (pts)		
		GSM8K	MMLU-Pro	HellaSwag
News zero-shot	Qwen-3.5-35B-MoE	0.0	0.0	2.4
News adapted	Gemma-4-31B	1.6	4.3	0.0
Reddit zero-shot	Minstral-3-14B	5.8	8.6	5.0
Reddit adapted	Gemma-4-31B	1.6	4.3	0.0
Hacker News zero-shot	Qwen-3.5-35B-MoE	0.0	0.0	2.4
Hacker News adapted	Gemma-4-31B	1.6	4.3	0.0
Math (arXiv) zero-shot	Qwen-3.5-35B-MoE	0.0	0.0	2.4
Math (arXiv) adapted	Qwen-3.5-35B-MoE	0.0	0.0	2.4

Table 7: What choosing on BPB costs you on each benchmark. For every corpus and reading, the model the ranking puts first and how many accuracy points that costs against each benchmark’s own top-1 model. Only three distinct picks occur in eight rows, which is itself the finding: the adapted ranking selects Gemma-4-31B on every general-text corpus—the HellaSwag leader outright, at 1.6 GSM8K and 4.3 MMLU-Pro points—while adapting on mathematics recovers the GSM8K and MMLU-Pro leader at zero regret. The corpus, not the metric, decides which column you pay in.

D Robustness and Measurement Artefacts

Stability across domains. All eleven models ran on three independently collected domains under the same recipe (Tables 4 and 12). Adaptation genuinely changes the choice—zero-shot and adapted rankings agree only at Spearman $\rho = 0.655$ on news, 0.609 on Reddit and 0.682 on Hacker News, with models moving up to seven places—and the adapted ordering then holds across domains: news and Reddit agree on nine of eleven exact ranks ($\rho = 0.991$, maximum displacement one), news against Hacker News gives 0.955 and Reddit against Hacker News 0.964, and the agreement is not merely ordinal (Figure 5). **So does the zero-shot ordering**—0.982, 0.964 and 0.973 on the same three comparisons. The tempting story is that adaptation buys stability, and it does not; it buys a *different* stable ordering, and Section 3.2 is the evidence that it is the better of the two.

Why one shared configuration. We first ran a per-model hyperparameter search and abandoned it because its budget was confounded with model size: inside a fixed wall-clock allocation the two largest models completed four trials each while the 1B models completed 17 and 18 (results/legacy_sweep/*/trials.csv). That hands small models a better-tuned adapter and biases the ranking by parameter count in a way no post-hoc correction can remove.

Model	Adapted BPB	MMLU-Pro	HellaSwag	GSM8K
Gemma-4-31B	0.624 (1)	57.1 (2)	87.9 (1)	85.0 (3)
Gemma-4-12B	0.653 (2)	44.8 (7)	84.0 (3)	64.8 (7)
Qwen-3.5-35B-MoE	0.678 (3)	61.4 (1)	85.6 (2)	86.6 (1)
Ministral-3-14B	0.679 (4)	52.8 (4)	82.9 (4)	80.8 (6)
Qwen-3.5-9B	0.708 (5)	55.6 (3)	81.6 (5)	85.8 (2)
Qwen-2.5-7B	0.717 (6)	47.2 (6)	80.3 (6)	81.3 (4)
Qwen-3.5-4B	0.749 (7)	52.4 (5)	77.2 (7)	81.0 (5)
Llama-3.2-1B	0.771 (8)	11.2 (11)	65.8 (9)	6.4 (11)
Qwen-2.5-1.5B	0.797 (9)	25.6 (8)	68.0 (8)	61.6 (8)
LiquidAI-LFM2.5	0.852 (10)	16.4 (9)	60.9 (10)	54.1 (9)
Qwen-2.5-0.5B	0.901 (11)	16.0 (10)	51.8 (11)	34.3 (10)

Table 8: Where the readings disagree, in one place. *Eleven-model cohort, news.* Adapted news BPB (lower is better) beside the three benchmark scores, rank in parentheses. Gemma-4-12B is the case to look at: second by BPB, third on HellaSwag, and *seventh* on both MMLU-Pro and GSM8K—and second by the independent in-domain criterion (Table 1). Llama-3.2-1B and Qwen-2.5-1.5B are adjacent in BPB and 2.1 HellaSwag points apart, yet 55 GSM8K points apart.

The cold start, seen a second way. Varying the inference context on 64 fixed held-out target windows from 16 to 2048 tokens reproduces Figure 3’s effect non-positionally: both models improve with context and the larger improves faster, so any protocol that scores short contexts under-rates the models that use long ones. Measured context sensitivity is 0.032 bits per byte per doubling, which is the constant the block-alignment bound below uses.

Token-aligned blocks: a real bias, bounded. Evaluation blocks are 512 *tokens* rather than a fixed byte span, so models with different tokenizers score slightly different amounts of text—5.92–6.24 MB across the eleven-model cohort, a 5% spread, and 9.9% across the seventeen. Bounding the resulting bias with the measured context sensitivity puts it at $0.08\times$ the smallest adjacent rank gap in the eleven-model cohort. In the extended cohort the bound is looser and worth stating: on the worst adjacent pair, Qwen2.5-3B over Falcon3-7B, whose BPB gap is only 0.29%, the bound reaches $0.61\times$ the gap. It still **overturns no rank**, but byte-aligned blocking would remove the question, and we recommend it for future work.

The resolved comparison is not an artifact of a lucky seed. Every adaptation cell ran with the `torch` RNG unseeded, and the cluster bootstrap resamples models, so it carries no within-cell training variance. A preregistered check—three seed replicates for all 17 models, 51 cells, news corpus, everything else pinned—is confirmatory on all four questions it asked. Median within-model SD of adapted BPB is 0.000087 against a between-model SD of 0.074340, a ratio of 0.0012; in-band pairwise accuracy is 0.900 under each seed; *no* in-band pair—0 of 40—reverses its ordering; and adding a seed-resampling stage moves the resolved interval by 0.001. The spread is not uniform: Gemma-4-12B is the one model where seed noise is visible at all (SD 0.00315, $5.3\times$ the next-largest spread).

Steering: direction, not magnitude. Adapting on general text raises agreement with HellaSwag on news from $\rho = 0.718$ to 0.982 while GSM8K falls from 0.800 to 0.700 and MMLU-Pro from 0.845 to 0.773; adapting on contamination-screened arXiv mathematics instead raises GSM8K agreement to 0.909 and restores the GSM8K oracle pick at zero regret, while three general-text corpora tried as controls all leave GSM8K at or below the do-nothing baseline. That is the rotation, and it is why the body reports only its direction. Because the steering hypothesis predicts a *joint* movement, the better-powered statistic is a difference in differences—(GSM8K minus HellaSwag selection accuracy under mathematics adaptation) minus the same quantity under a general-text corpus—which is $+0.242$ pooled. Resampling models gives a strictly positive difference in 93.7% of draws; the stricter publisher-family clustering the mechanism itself implies gives 80.9% with not one negative draw in 20,000, but no longer clears 0.05 on any tie convention. A leave-one-family-out jackknife shows why: dropping the two Gemma models takes the pooled difference from $+0.242$ to $+0.056$. The remedy is more *families*, not more models.

Discarding the cold start: how much does it buy? (E10.) Each model was re-scored zero-shot with per-token loss accumulated separately at every position in the 512-token block, so every cutoff K is derivable from one pass; the falsifying outcome was fixed in advance—if any K brought the free reading within one in-band pair of adapted BPB, the mechanism claim above would have been withdrawn. It does not. Under lenient scoring in-band accuracy rises from 0.545 at $K=0$ to 0.636 at $K=128$ and no further, closing 25% of the 0.364 gap to adapted BPB. Under strict scoring the test is *uninformative* and we say so rather than counting it: the two readings already tie at 0.727 there, so there is no gap for the cold start to be accused of causing, and discarding positions only makes the free reading worse (0.636 at $K \geq 128$). The re-measured cells are not the published ones—they run up to 1.5% low—so every cutoff is compared against this run’s *own* $K=0$, which reproduces the published zero-shot selector exactly under both conventions.

Scoring	Relative BPB gap	All in-band pairs		Criterion-resolved only	
		pairs	agreement	pairs	agreement
lenient	all	40	0.900	35	0.971
	$\geq 1\%$	38	0.947	35	0.971
	$\geq 2\%$	33	1.000	32	1.000
	$\geq 5\%$	22	1.000	22	1.000
strict	all	40	0.850	31	0.968
	$\geq 1\%$	38	0.895	31	0.968
	$\geq 2\%$	33	0.909	28	0.964
	$\geq 5\%$	22	0.955	19	1.000

Table 9: How often the lower adapted-BPB model is the better model, by how far apart they are. *Extended cohort: 17 models, 40 size-matched pairs.* Gaps are a percentage of the smaller adapted BPB. The measurement’s own floor is about 1.4% (seed variation plus block alignment); Section 4 rounds its rule up to the 2% row, where these counts begin.

Three reproduction controls. Re-running two Ministral-3-14B cells weeks later on different hardware reproduced zero-shot BPB exactly and adapted BPB to within 0.0003, two orders of magnitude below the differences compared. Because the extension models were adapted on a different software stack, a published cell (Qwen2.5-7B) was re-run through the extension pipeline; the analysis refuses to pool the cohorts unless it reproduces within 1%, and it reproduces to 0.01% on adapted BPB (0.7169 against 0.7170). And because benchmark rankings are themselves estimates, recomputing selection accuracy while discarding pairs whose benchmark gap is smaller than k combined standard errors leaves the result intact and slightly improves it: on GSM8K 0.909 at $k=0$ becomes 0.933 at $k=3$, and on HellaSwag 0.836 becomes 0.863, so the metric’s errors concentrate in pairs the benchmark itself cannot separate.

E The Downstream Fine-Tune

Section 3.4 reports the result; this is how it was built, and what it does and does not license. It ran in two stages, each preregistered in code before its job returned: first the two pairs on which the selectors disagree most (Gemma-4-12B against Ministral-3-14B, Llama-3.2-1B against Qwen-2.5-1.5B), then every cohort model that fits one GPU, so that no pair is chosen by anyone. Gemma-4-31B and Qwen-3.5-35B-MoE need more than one GPU *for this fine-tune*—its 1,280-token sequences and its generation pass cost far more memory than the 512-token adaptation priced in Section 1—and are excluded by that rule, which costs one in-band pair of forty.

The task. Journalistic convention makes a news article’s first paragraph a standalone summary of it, so the corpus supplies its own supervision: the input is the article with its opening paragraph removed, truncated to 3,000 characters, and the target is that paragraph. Articles qualify if they have at least three paragraphs, an opening paragraph of 200–700 characters ending in sentence punctuation, and at least 800 characters of remaining body. Of the 11,905 validation documents, 3,627 qualify; 2,000 are sampled for training and 500 for evaluation under a fixed seed.

Why the validation split. The in-domain criterion (Section 2.2) and the BPB test blocks both live in the *test* split. That is verified rather than assumed: the shipped criterion items reproduce item-for-item from an independent reconstruction of the test split, and the adaptation pipeline’s contamination pass—which would have split the corpus differently—is off by default and recorded as off in every published cell. Evaluation documents are then hashed against the test documents twice, on full normalised text and on a 400-character opening, the second because a syndicated wire story is republished with an edited body and an identical lede. Four articles were dropped by that second check and none by the first.

Equal preparation. Both members of a pair get the paper’s own adaptation recipe unchanged —LoRA rank 16, $\alpha=32$, dropout 0.05, learning rate 10^{-4} , effective batch 32, 250 steps—with prompt tokens masked out of the loss so the model is trained to write the lede rather than to reconstruct the article. Nothing is tuned on either model, because a per-model search is exactly the cost this protocol exists to avoid. The tokenizers differ, so the budget is matched in update steps $\times$ batch $\times$ sequence length; each model’s realised token count is in the artifact. Decoding is greedy with a 180-token cap and each checkpoint’s own generation configuration otherwise untouched, following Appendix G. The prompt is assembled so that the instruction always survives truncation: the *body* is cut to fit, never the instruction after it, because right-truncating the assembled prompt would delete the instruction from the longest articles at a different article count for each tokenizer—an asymmetry between the two models being compared. At most four of the 500 evaluation articles were cut for any model, and none for three of them.

The decision rules, fixed before each run. For the two chosen pairs: mean ROUGE-L F1 over the 500 held-out articles, a paired bootstrap over articles (20,000 draws, 95% percentile interval) and a practical-equivalence margin of 0.01—an interval excluding zero resolves the pair for whichever reading predicted that winner. The first pair resolved for adapted BPB; the second was inconclusive. For the full cohort: the fine-tuned ROUGE-L is treated as the criterion and every selector is scored with the paper’s own statistic—pairwise accuracy within $2\times$, model-level cluster bootstrap, 20,000 draws (Table 10). A separate guard voids any comparison in which fine-tuning failed to improve a model over its own base weights, since ordering unmoved models would order their base behaviour under another name. It did not fire: all fifteen models gained between +0.041 and +0.153.

A second recipe. Both readers of an earlier draft raised the same objection: the fine-tune reused the selector’s own recipe, so a model that adapts quickly under it could win both. Recipe B changes every setting the two shared—LoRA rank 64, $\alpha=128$, learning rate 3×10^{-4} , effective batch 16, 400 steps, a new seed—on the same task, data, decoding and metric, preregistered before it ran. It fits the training articles far harder and scores slightly below recipe A for every model, yet orders the fifteen systems almost identically (Spearman 0.982). Against recipe-B systems adapted BPB remains the top selector at 0.923 [0.750, 1.000], ahead of HellaSwag at 0.846, parameter count at 0.744 and GSM8K and MMLU-Pro near chance, and it picks the better system on all 20 pairs recipe B resolves. Its margin over parameter count, resolved under recipe A, does not resolve here (+0.179 [−0.036, +0.441]), so that margin should be read as recipe-specific (Table 10).

What copying buys. A news body restates its own lede, so reference overlap is partly available by retrieval. Two reference points bound it: the article’s first body sentence scores 0.125 against the reference and a random body sentence 0.111. Unadapted base weights land close to copying—inspection confirms they largely do copy—while every fine-tuned system clears it, from 0.188 (Qwen-2.5-0.5B) to 0.249 (Gemma-4-12B).

How close the corpus has to be. Exploratory, run after the result at a reader’s request: scoring the news fine-tune with adapted BPB from each of the other corpora, on the ten in-band pairs of the nine fine-tuned models that have all four, gives 0.900 for news and for Reddit, 0.700 for Hacker News and 0.600 for arXiv mathematics; the unadapted readings reach 0.500–0.600. Fresh general text does most of the work and nearby text does it best. Ten pairs support no interval, so this is a direction, not a result.

What this does not show. One task, one domain, two recipes, one seed each. On this task adapted BPB ordered the fine-tuned systems better than any other selector and GSM8K and MMLU-Pro ordered them at chance; its margin over HellaSwag does not resolve. It is not a claim that adapted BPB predicts downstream utility in general, and the paper does not make one.

	All 39 pairs	95% CI	22 resolved pairs	Recipe B
Adapted BPB	0.949	[0.817, 1.000]	0.955	0.923
HellaSwag	0.872	[0.640, 1.000]	0.909	0.846
Zero-shot BPB	0.769	[0.444, 1.000]	0.682	0.744
Parameter count	0.718	[0.463, 0.912]	0.773	0.744
MMLU-Pro	0.513	[0.250, 0.846]	0.500	0.538
GSM8K	0.474	[0.194, 0.824]	0.455	0.447

Table 10: Which selector predicts the fine-tuned system? Fifteen models, each fine-tuned identically on the lede task; the criterion is each system’s ROUGE-L on 500 held-out articles. Each row is a selector’s pairwise accuracy over every size-matched pair, with the model-level cluster bootstrap, and its accuracy on the 22 pairs whose fine-tuned difference is itself resolved by a paired bootstrap over articles. The last column scores the same selectors against systems built with the deliberately different recipe B. GSM8K ties one pair and is scored on 38.

F The Cohorts

Table 11 is the extended 17-model cohort behind Figure 1 and Section 3.1: every model, its parameter count, its adapted news BPB and its score on the in-domain criterion with that criterion’s own interval. Table 12 then gives the original eleven across all three general corpora and both BPB endpoints, which is the population every other analysis in the paper uses. Table 12 is the full three-corpus leaderboard: both endpoints for all eleven models on news, Reddit

Model	Params (B)	Adapted BPB	In-domain criterion		Cohort
			score	95% CI	
Gemma-4-31B	31.4	0.624	0.213	[0.197, 0.229]	original
Mistral-Nemo-12B	12.3	0.646	0.166	[0.152, 0.181]	added
Gemma-4-12B	12.0	0.653	0.186	[0.171, 0.201]	original
Qwen-2.5-14B	14.8	0.675	0.137	[0.124, 0.151]	added
Qwen-3.5-35B-MoE*	34.7	0.678	0.147	[0.134, 0.161]	original
Ministral-3-14B	14.0	0.679	0.142	[0.128, 0.155]	original
Qwen-3.5-9B	9.0	0.708	0.116	[0.104, 0.129]	original
Qwen-2.5-7B	7.7	0.717	0.121	[0.108, 0.134]	original
Falcon3-10B	10.4	0.728	0.094	[0.083, 0.106]	added
Qwen-3.5-4B	4.2	0.749	0.101	[0.089, 0.112]	original
Qwen-2.5-3B	3.1	0.756	0.107	[0.095, 0.119]	added
Falcon3-7B	7.5	0.758	0.086	[0.075, 0.097]	added
Llama-3.2-1B	1.2	0.771	0.104	[0.092, 0.116]	original
SmolLM2-1.7B	1.7	0.788	0.091	[0.080, 0.102]	added
Qwen-2.5-1.5B	1.6	0.797	0.089	[0.078, 0.100]	original
LiquidAI-LFM2.5	1.2	0.852	0.078	[0.067, 0.088]	original
Qwen-2.5-0.5B	0.5	0.901	0.066	[0.057, 0.076]	original

Table 11: The extended cohort, in full. All 17 models ordered by adapted news BPB, with the criterion score each is judged against and its paired item-bootstrap interval. “added” marks the six models admitted for the size-matched extension (Section 3.1); the other eleven carry the mechanism, four-corpus and stability analyses. *Qwen-3.5-35B-MoE activates 3B of its 34.7B parameters; parameter count here is always the total.

and Hacker News, ordered by adapted news BPB. It is the artifact every eleven-model number in the paper derives from, as Table 11 is for the seventeen.

The study ran at eleven models before the extension, which supersedes those numbers without contradicting them: selection accuracy exceeded chance in 23 of the 24 (corpus, reading, benchmark) cells—strongest for adapted BPB against HellaSwag on Hacker News (0.964, CI [0.840, 1.000]) and on news (0.964, CI [0.875, 1.000]), with mathematics-adapted BPB against GSM8K at 0.891 [0.725, 1.000], the one exception being mathematics-corpus *zero-shot* BPB against HellaSwag (0.727, CI [0.478, 0.980]). Between-corpus contrasts did not separate (+0.127

Model	News		Reddit		Hacker News	
	zero-shot	adapted	zero-shot	adapted	zero-shot	adapted
Gemma-4-31B	0.752	0.624	1.062	0.913	0.962	0.805
Gemma-4-12B	0.866	0.653	1.172	0.942	1.063	0.844
Qwen-3.5-35B-MoE	0.707	0.678	1.008	0.963	0.866	0.836
Ministral-3-14B	0.718	0.679	1.008	0.947	0.876	0.835
Qwen-3.5-9B	0.740	0.708	1.039	0.983	0.894	0.860
Qwen-2.5-7B	0.758	0.717	1.051	0.991	0.922	0.882
Qwen-3.5-4B	0.782	0.749	1.080	1.020	0.933	0.898
Llama-3.2-1B	0.811	0.771	1.109	1.055	1.011	0.976
Qwen-2.5-1.5B	0.833	0.797	1.128	1.072	1.002	0.967
LiquidAI-LFM2.5	1.340	0.852	1.779	1.149	1.687	1.051
Qwen-2.5-0.5B	0.935	0.901	1.222	1.161	1.099	1.064

Table 12: Marker-masked BPB under one fixed adaptation protocol. Lower is better. Rows are ordered by adapted news BPB; all eleven models occur in all three corpora.

on GSM8K and MMLU-Pro, -0.127 on HellaSwag, every interval containing zero), and adapted BPB was the only in-band selector clearing the in-domain criterion (0.909, against 0.818 HellaSwag, 0.636 parameter count and 0.545 unadapted). Eleven models give eleven in-band pairs, and at that size a selector must be near-perfect to separate—which is what the extension in Section 3.1 fixed, and why the earlier claim that no benchmark clears chance was a statement about power rather than about benchmarks. Three checkpoints are excluded. The two OLMo-2 models fell under the preregistered infrastructure-fault rule: their adapted BPB cells exist and are unused, because the criterion’s generation engine would not initialise for them, identically before and after a cache rebuild. The third, Qwen-2.5-32B, did not finish its 250 steps inside the run window; it would have added about two in-band pairs at the top of the size range. The OLMo exclusion removed a whole publisher, which is what the cohort most needs.

G The Benchmark Column Re-Score

Harness configuration. MMLU-Pro is 5-shot chain-of-thought with exact match on one stratified 1,000-question subset (seed 42); HellaSwag is 10-shot length-normalised multiple-choice accuracy on all 10,042 validation examples; GSM8K is 5-shot chain-of-thought with flexible numeric extraction on all 1,319 test examples. All bf16, greedy, tokenizer-native BOS.

July’s eleven-model column was scored on a host that no longer exists and generative scores do not pool across hosts, so it was re-scored in full—seventeen models $\times$ three benchmarks = fifty-one cells, one host (A100-SXM-64GB), one build—with the design and the exclusion rule fixed in advance. Task definitions match exactly (17/17 identical task hashes on both hashed benchmarks) and every overlapping cell agrees with July to within 1.3 points, most to within 0.3.

Reaching that took one correction worth reporting, because the fault was in our own fix. Because the newer backend honours each checkpoint’s shipped `generation_config.json` and the original did not, we replaced every checkpoint’s generation settings with one identical policy—which also discarded `suppress_tokens`, a setting that is live for gemma-4-12B. That single omission cost it 9.86 GSM8K and 14.30 MMLU-Pro points while leaving HellaSwag, scored by likelihood and never calling `generate()`, intact. Keeping the setting for all seventeen models reproduces July to $+1.21$ points on GSM8K, $+0.10$ on MMLU-Pro and -0.27 on HellaSwag. That GSM8K cell is the one outside the preregistered 1.0-point tolerance, within the allowance of one that the re-score declared in advance: the column is within its stated allowance rather than clean, and those are different claims. The lesson generalises: a wrapper imposed to remove a confound can be the confound, so the unmodified harness belongs in the design as a control.